Topic 9 — Time Series: White Noise, AR, MA, ARMA and ARIMA Models

What Is a Time Series?

Definition

A **time series** is an infinite sequence of random numbers parametrized (indexed) by (discrete) time: $X_1, X_2, X_3, \ldots, X_n, \ldots \in \mathbb{R}$. In real data, the values $X_1, X_2, \ldots$ are typically **not independent**.

Example: $X_n = 1.005\, X_{n-1} + Y_n$, where Y_n is random and may itself depend on time n.

Stationarity

Definition

A time series $\{X_n\}$ is called **(weakly) stationary** if:

1. $\mathbb{E}(X_n)$ does not depend on n (constant);
2. $\mathbb{E}(X_n^2) < \infty$ for all n;
3. $\operatorname{cov}(X_n, X_{n+m}) = \mathbb{E}(X_n X_{n+m}) - \mathbb{E}(X_n) \cdot \mathbb{E}(X_{n+m})$ does **not** depend on n — it depends on m only.

Remark

In other words, $\operatorname{cov}(X_n, X_\ell)$ depends only on the time-lag $\ell - n$. If $\{X_n\}$ is stationary, then $\operatorname{Var}(X_n) = \operatorname{cov}(X_n, X_n)$ does not depend on n either, and neither does $\operatorname{corr}(X_n, X_\ell)$ (as a function of $\ell - n$).

Example: A Stationary Time Series

Example

Let Y, Z be uncorrelated, $\mathbb{E}(Y) = \mathbb{E}(Z) = 0$, $\mathrm{Var}(Y) = \mathrm{Var}(Z) = \sigma^2$ (e.g. independent $N(0, \sigma^2)$). For fixed $\lambda \in [0, 2\pi]$, define

$$X_n = Y\cos(n\lambda) + Z\sin(n\lambda).$$

Then $\mathbb{E}(X_n) = \cos(n\lambda)\mathbb{E}(Y) + \sin(n\lambda)\mathbb{E}(Z) = 0$ — constant in n.

Example

Using $\mathbb{E}(Y^2) = \mathbb{E}(Z^2) = \sigma^2$ and $\mathbb{E}(YZ) = \mathrm{cov}(Y, Z) = 0$, a direct expansion gives

$$\mathbb{E}(X_n X_{n+m}) = \big(\cos(n\lambda)\cos((n+m)\lambda) + \sin(n\lambda)\sin((n+m)\lambda)\big)\sigma^2 = \cos(m\lambda)\,\sigma^2.$$

So $\mathbb{E}(X_n^2) = \sigma^2 < \infty$ and $\mathrm{cov}(X_n, X_{n+m}) = \cos(m\lambda)\sigma^2$ depends on m only: $\{X_n\}$ is **stationary**.

White Noise

Definition

A time series $\{Z_n\}$ is a **white noise** if (1) $\mathbb{E}(Z_n) = 0$, and (2) for some $\sigma > 0$,

$$\operatorname{cov}(Z_n, Z_{n+m}) = \begin{cases} \sigma^2, & m = 0, \\ 0, & m \neq 0. \end{cases}$$

I.e. white noise has zero mean and uncorrelated values.

Remark

The standard example of white noise: i.i.d. $Z_n \sim \mathcal{N}(0, \sigma^2)$.

Autoregressive Model AR(1)

Example

$$X_n = \mu + \alpha(X_{n-1} - \mu) + Z_n, \qquad X_0 = a \in \mathbb{R},$$

where μ, α are constants and $\{Z_n\}$ is white noise, independent of $\{X_n\}$. Then $X_1 = \mu + \alpha(a - \mu) + Z_1$, $X_2 = \mu + \alpha(X_1 - \mu) + Z_2$, and so on — each new value uses a fresh, independent Z_n.

Writing $Y_n := X_n - \mu$ gives $Y_n = \alpha Y_{n-1} + Z_n$. We look for conditions making $\{X_n\}$ (hence $\{Y_n\}$) stationary.

AR(1): Finding the Stationarity Condition

If $\mathbb{E}(Y_n) = k$ (constant), then $k = \alpha k + 0$, so $\alpha = 1$ or $k = 0$. Since $\alpha = \pm 1$ would force $\sigma = 0$ (impossible), we need $\alpha \neq \pm 1$ and $k = 0$, i.e. $\mathbb{E}(X_n) = \mu$.

If $\mathrm{Var}(Y_n) = v$ (constant), independence of Z_n gives $v = \alpha^2 v + \sigma^2$, i.e. $v(1 - \alpha^2) = \sigma^2$, so

$$\mathrm{Var}(X_n) = v = \frac{\sigma^2}{1 - \alpha^2}.$$

Since $\mathrm{Var}(X_n) \geq 0$, we need $1 - \alpha^2 > 0$, i.e. $|\alpha| < 1$.

Definition

$|\alpha| < 1$ is necessary and sufficient for stationarity of $X_n = \alpha X_{n-1} + Z_n$. Equivalently, using the **characteristic equation** $1 - \alpha\lambda = 0$ (so $\lambda = \frac{1}{\alpha}$): the series is stationary iff $|\lambda| > 1$.

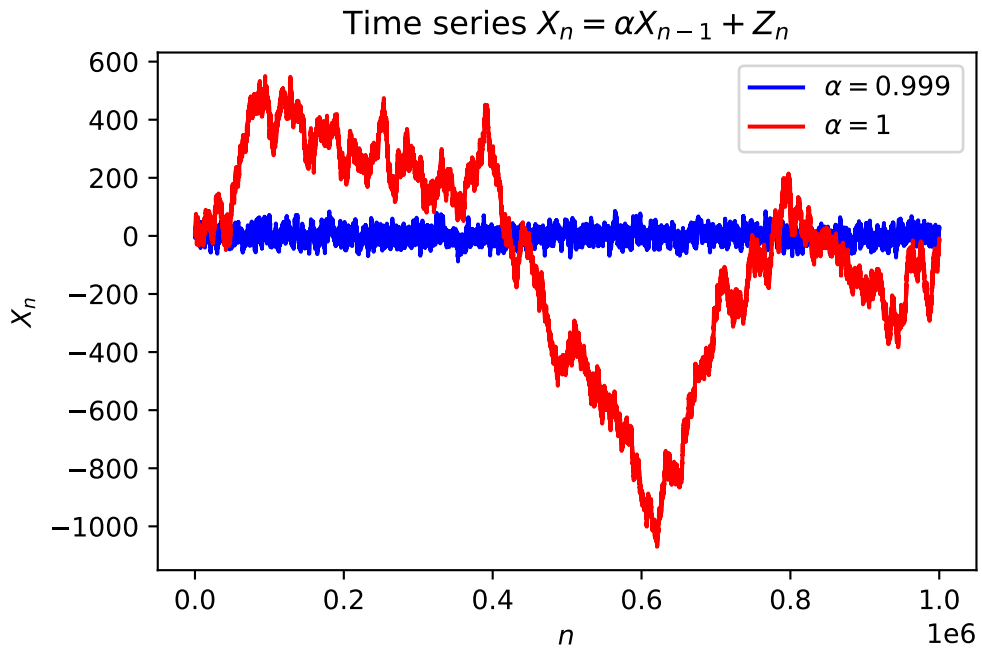


Figure 1: Qualitative difference between stationary (blue) and non-stationary (red) time series behaviour

Autoregressive Model AR(2)

Example

$$X_n = \mu + \alpha_1(X_{n-1} - \mu) + \alpha_2(X_{n-2} - \mu) + Z_n, \qquad \alpha_2 \neq 0,$$

with initial values $X_0 = a$, $X_1 = b$. Then $X_2 = \mu + \alpha_1(b - \mu) + \alpha_2(a - \mu) + Z_2$, then X_3 uses X_2 and X_1, and so on. With $Y_n = X_n - \mu$: $Y_n = \alpha_1 Y_{n-1} + \alpha_2 Y_{n-2} + Z_n$.

Definition

The **characteristic equation** is $1 - \alpha_1\lambda - \alpha_2\lambda^2 = 0$, with two roots λ_1, λ_2 (possibly complex). The time series is **stationary** iff $|\lambda_1| > 1$ **and** $|\lambda_2| > 1$.

Reminder: Quadratic Equations and Complex Numbers

Remark

$a\lambda^2 + b\lambda + c = 0$ $(a \neq 0)$ has roots $\lambda_{1,2} = \dfrac{-b \pm \sqrt{D}}{2a}$, discriminant $D = b^2 - 4ac$. If $D < 0$, the roots are complex: $\lambda_{1,2} = p \pm qi$ $(i^2 = -1)$, with $|p \pm qi| = \sqrt{p^2 + q^2}$.

Example 1: AR(2), Stationary

Example

$$X_n = \tfrac{1}{12}X_{n-1} + \tfrac{1}{2}X_{n-2} + Z_n.$$

Characteristic equation: $\lambda^2 + \tfrac{1}{6}\lambda - 2 = 0$, $D = \tfrac{1}{36} + 8 = \tfrac{289}{36}$, $\sqrt{D} = \tfrac{17}{6}$.

$$\lambda_1 = -\tfrac{3}{2}, \qquad \lambda_2 = \tfrac{4}{3}.$$

Both $|\lambda_1|, |\lambda_2| > 1$: **stationary**.

Example 2: AR(2), Non-Stationary

Example

$$X_n = \tfrac{1}{3}X_{n-1} + \tfrac{2}{3}X_{n-2} + Z_n.$$

Characteristic equation: $2\lambda^2 + \lambda - 3 = 0$, $D = 25$.

$$\lambda_1 = -\tfrac{3}{2}, \qquad \lambda_2 = 1.$$

$|\lambda_1| > 1$ but $|\lambda_2| = 1 \not> 1$: **not stationary**.

Example 3: AR(2), Complex Roots

Example

$$X_n = 2X_{n-1} - 2X_{n-2} + Z_n.$$

Characteristic equation: $2\lambda^2 - 2\lambda + 1 = 0$, $D = 4 - 8 = -4 < 0$ (complex roots):

$$\lambda_{1,2} = \frac{2 \pm \sqrt{-4}}{4} = \frac{1}{2} \pm \frac{1}{2}i.$$

$$|\lambda_{1,2}| = \sqrt{\left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{2}\right)^2} = \frac{\sqrt{2}}{2} < 1.$$

Not stationary.

Autoregressive Model AR(p)

$$X_n = \mu + \alpha_1(X_{n-1} - \mu) + \cdots + \alpha_p(X_{n-p} - \mu) + Z_n.$$

Example

E.g. AR(3): $X_n = aX_{n-1} + bX_{n-2} + cX_{n-3} + Z_n$. Given a, b, c, X_0, X_1, X_2, generate Z_3 to get $X_3 = aX_2 + bX_1 + cX_0 + Z_3$, then $X_4 = aX_3 + bX_2 + cX_1 + Z_4$, and so on.

Definition

The characteristic equation $1 - \alpha_1\lambda - \cdots - \alpha_p\lambda^p = 0$ has p roots $\lambda_1, \ldots, \lambda_p$ (possibly complex). The AR(p) series is stationary iff $|\lambda_1| > 1, \ldots, |\lambda_p| > 1$.

The roots may be found numerically (e.g. in Python) for larger p.

Moving Averages MA(q)

Definition

$$X_n = \mu + Z_n + \beta_1 Z_{n-1} + \cdots + \beta_q Z_{n-q}.$$

Since $\mathbb{E}(Z_{n-j}) = 0$ for all j, $\mathbb{E}(X_n) = \mu$ — constant. To check the covariance condition, take $q = 1$, $\mu = 0$: $X_n = Z_n + \beta Z_{n-1}$, $X_{n+m} = Z_{n+m} + \beta Z_{n+m-1}$ $(m > 0)$. Expanding,

$$\operatorname{cov}(X_n, X_{n+m}) = \underbrace{\operatorname{cov}(Z_n, Z_{n+m})}_{=0} + \beta \underbrace{\operatorname{cov}(Z_{n-1}, Z_{n+m})}_{=0} + \beta \underbrace{\operatorname{cov}(Z_n, Z_{n+m-1})}_{=\sigma^2 \text{ if } m=1,\ 0 \text{ if } m>1} + \beta^2 \underbrace{\operatorname{cov}(Z_{n-1}, Z_{n+m-1})}_{=0}$$

— depends on m only.

Remark

In other words, MA(q) is always stationary.

Full ARMA(p, q) Model

Definition

"AR" = autoregressive, "MA" = moving average — this model includes past white noise too:

$$X_n = \mu + \alpha_1(X_{n-1} - \mu) + \cdots + \alpha_p(X_{n-p} - \mu) + Z_n + \beta_1 Z_{n-1} + \cdots + \beta_q Z_{n-q}.$$

Remark

The characteristic equation and stationarity condition for ARMA(p, q) coincide with those of its AR(p) component — **you may ignore the** β**'s** for stationarity.

The ARIMA(p, d, q) Model: Motivation

Consider an ARMA(2,q) model with characteristic equation $1 = \alpha_1\lambda + \alpha_2\lambda^2$. Suppose $\lambda = 1$ is a root: then $|\lambda| \not> 1$, so $\{X_n\}$ is **not stationary**.

Substituting $\lambda = 1$: $1 = \alpha_1 + \alpha_2$, so $\alpha_1 = 1 - \alpha_2$. Then

$$X_n = \alpha_1 X_{n-1} + \alpha_2 X_{n-2} + Z_n + \cdots = (X_{n-1} - \alpha_2 X_{n-1}) + \alpha_2 X_{n-2} + Z_n + \cdots$$

$$X_n - X_{n-1} = -\alpha_2(X_{n-1} - X_{n-2}) + Z_n + \cdots.$$

Denoting $Y_n := X_n - X_{n-1}$ (the **differencing** of X_n): $Y_n = -\alpha_2 Y_{n-1} + Z_n + \cdots$ — an ARMA(1,q) series. If $|\alpha_2| < 1$, Y_n is stationary.

ARIMA(p, d, q): Definition

Definition

If the characteristic equation of an ARMA(2,q) model has roots $\lambda_1 = 1$ and $|\lambda_2| > 1$, then $\{X_n\}$ is not stationary, but $Y_n = X_n - X_{n-1}$ is. We then say $\{X_n\}$ follows an **ARIMA(1,1,q)** model — the first "1" because Y_n is ARMA(1,0)-like, the second "1" because differencing was applied once.

Recovering X_n from Y_n:

$$Y_n = X_n - X_{n-1} \iff X_n = Y_n + X_{n-1} = Y_n + Y_{n-1} + X_{n-2} = \cdots = Y_n + Y_{n-1} + \cdots + Y_1 + X_0.$$